Chapter 10 – Multimedia and Web

GE-161 Introduction to Information and Communication Technologies

Prof. Dr. Syed Waqar ul Qounain
Department of Information Technology
University of the Punjab, Lahore

Learning Objectives

1. Define Web-based multimedia and list some advantages and disadvantages of using multimedia.
2. Describe each of the following multimedia elements—text, images, animation, audio, and video—and tell how they differ.
3. Briefly describe the basic steps and principles involved with designing a multimedia Web site.
4. List the various tasks involved with developing a multimedia Web site.
5. Explain how markup languages, scripting languages, and other tools are used today to create multimedia Web pages.
6. Discuss the possible format of Web-based multimedia in the future.

Overview

- This chapter covers:
  - What Web-based multimedia is and how it is used today
  - The advantages and disadvantages of using multimedia
  - A look at basic multimedia elements
  - Steps and principles in designing a multimedia site
  - How a multimedia Web site is developed and the software used during this process
  - A look at the future of Web-based multimedia

What Is Web-Based Multimedia?

- Multimedia: The integration of a variety of media, such as text, images, video, animation, and sound
- Web-based multimedia (also called rich media): Multimedia located on Web pages
- Multimedia sites often contain elements that users interact with directly
  - Control the delivery of a sound or video clip, manipulate a 3D object, play a game, etc.
- Fast computers and broadband Internet connections make Web-based multimedia much more feasible than in the past
- Vast majority of Web sites today include multimedia (advertisements, TV shows, podcasts, user generated content)

Web-Based Multimedia Applications

- Information delivery: Photos of products, video clips, animation to convey concepts, etc.
- E-commerce: Photos of products, samples of movies and music, etc.
  - Virtual reality (VR): The use of a computer to create three-dimensional environments that look like they do in the real world. (homes for sale, etc.)
  - Augmented reality: Overlaying computer generated images on top of real time images
- Entertainment: Online TV/movies, music, games, etc.

Web-Based Multimedia Applications

FIGURE 10-1
Web-based multimedia applications.

INFORMATION DELIVERY
Images, video, audio, and animation are often used to convey information, such as via this Web-based training activity.

E-COMMERCE
Images and video are often used for e-commerce purposes, such as this virtual model home tour located on a home builder's Web site.

ENTERTAINMENT
Video and audio are often used in entertainment applications, such as this site that offers TV shows for online viewing.

3D VIRTUAL WORLDS
Images, animation, and sound are integral components of many 3D virtual worlds, such as Second Life shown here.

Advantages and Disadvantages of Using Web-Based Multimedia

- Advantages:
  - Can address a variety of learning styles
    - Visual learners
    - Auditory learners
    - Kinesthetic learners
  - Material more interesting and enjoyable
  - Many ideas are easier to convey
- Disadvantages:
  - Time and cost of development
  - Compatibility and download time for Web-based multimedia

Advantages of Using Web-Based Multimedia

MULTIMEDIA INSTRUCTIONS

TEXT-ONLY INSTRUCTIONS

FIGURE 10-2
Multimedia-based applications are often more effective than their single-medium counterparts.

Multimedia Elements

- Text: Used to supply basic content, and to add text-based menus and hyperlinks
  - Serif typefaces: More readable, used for large bodies of text
  - Sans serif typefaces: Used for titles, headings, Web page banners
  - Different typefaces can convey widely different feelings
  - Important to select a typeface that matches the style of the Web site
  - When a consistent text appearance is required—such as a logo—a graphical image is used instead

FIGURE 10-3

Typefaces. Typefaces are collections of text characters that share a common design and can be either serif or sans serif.

Multimedia Elements

- Images or graphics: Digital representations of photographs, drawings, charts, and other visual images also called a graphic
  - Clip art consists of pre-drawn electronic images
  - Stock photos are also available online

CLIP ART IMAGES
Typically use the GIF format and can be downloaded from a variety of Web sites. The images on this site are free for both personal and commercial use.

STOCK PHOTOGRAPHS
Typically use the JPEG format and are available through stock photograph agencies. The agency shown here has a variety of images organized by topics; all images require a fee for use, but all are royalty free.

FIGURE 10-4 Both clip art and stock photographs are plentiful on the Web.

Multimedia Elements

- GIF images: Commonly used for Web page line art images (logos, buttons, etc.)
  - 256 colors max
  - Use lossless file compression
  - Can be transparent
  - Can be interlaced

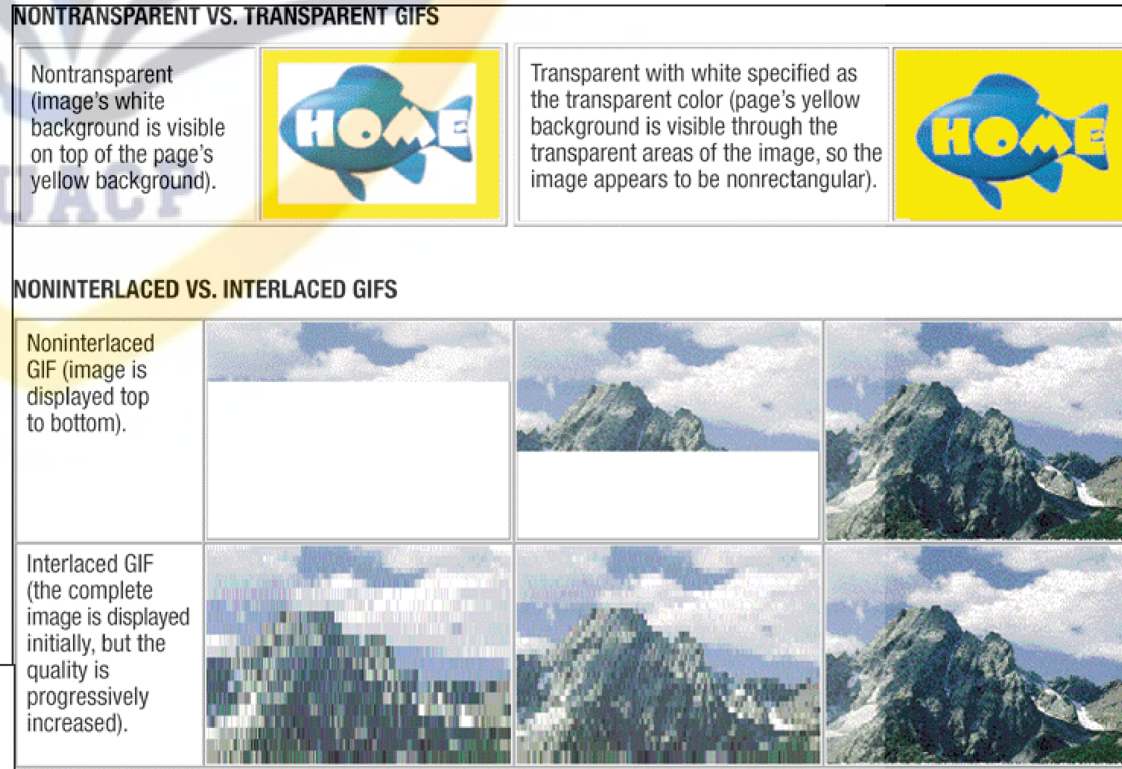


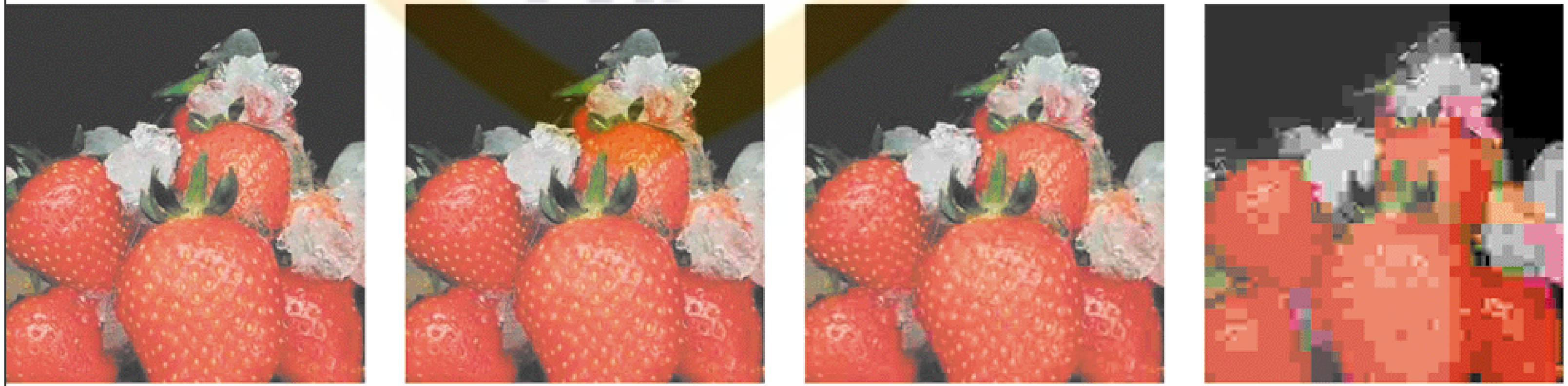
FIGURE 10-5
Transparent and interlaced GIFs.

Multimedia Elements

- PNG images: Format designed specifically for use with Web page images
  - Lossless compression, and with more efficiency than GIF
  - Can use color palette or true color
  - Can also be transparent and interlaced
- JPEG images: Commonly used for Web page photos
  - Uses lossy file compression
  - True color images
  - Can be progressive
  - The amount of compression is specified when the file is saved

Multimedia Elements

FIGURE 10-6
The amount of compression in a JPEG file affects both the file size and the display quality.

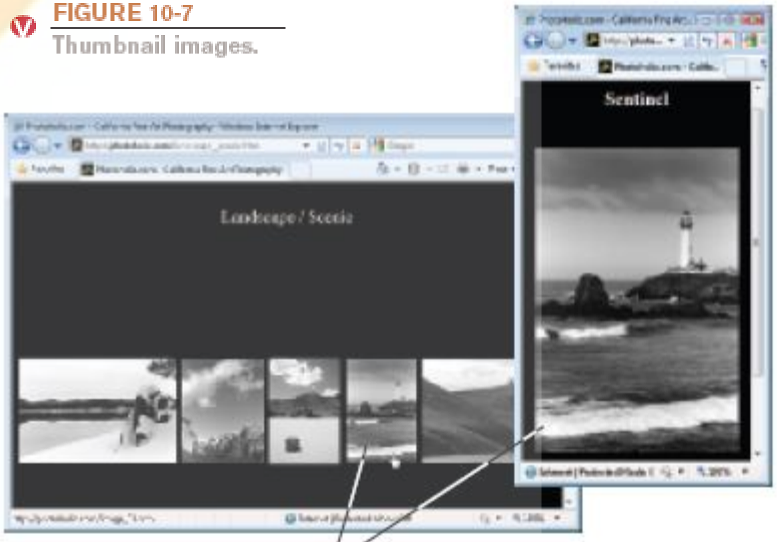


No compression (37 KB) | 40% compression (13 KB) | 80% compression (7 KB) | 100% compression (3 KB)

Multimedia Elements

- Choosing a graphic format
  - GIF or PNG—typically used for line art (clip art, logos, navigation buttons, etc.)
  - JPEG—typically used for photographs
  - Select the most appropriate format
  - For JPEG, balance compression quality and file size
  - Use thumbnail images

FIGURE 10-7
Thumbnail images.



Multimedia Elements

- Animation: A series of graphical images are displayed in succession to simulate movement
  - Java applet: A small program inserted into a Web page that performs a specific task
  - Animated GIF: A group of GIF images saved as an animated GIF file, inserted in a Web page, which are displayed successively to simulate movement

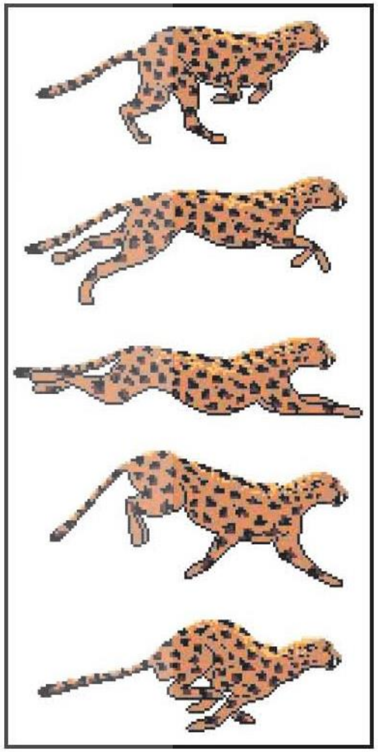
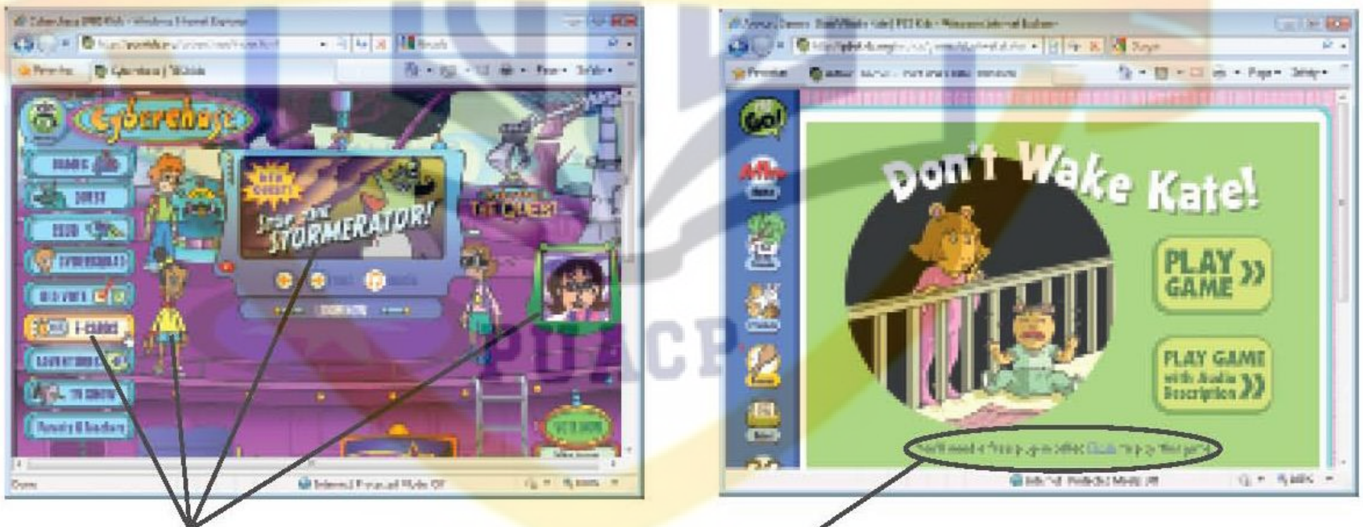


FIGURE 10-8
Animated GIFs.
When the animated GIF containing the images shown above are displayed one after the other, it appears that the leopard is running.

Multimedia Elements

– For more complex animations, developers can use JavaScript or another scripting language
  - Flash is in particularly wide use today
  - Silverlight is an alternative to Flash
– Many Wed-based animations require a plug-in
– Programming languages can be used to create multimedia elements and interactivity

Multimedia Elements



JAVASCRIPT
JavaScript is used in conjunction with Flash to light up the menu buttons as they are pointed to, to animate the characters, and to randomly change the images displayed in the two TV screens.

FLASH
Many games and other animated or interactive activities found on Web sites are created with Flash or Shockwave, so a plug-in is required to view the content.

FIGURE 10-9
JavaScript and Flash are commonly used on Web pages.

Multimedia Elements

- Audio: All types of sound including music, spoken voice, sound effects
  - Can be recorded using a microphone or MIDI instrument, captured from CDs, or downloaded from the Internet
  - Often played when an event occurs on a Web page or when the visitor clicks a link
  - Streaming audio is used to speed up delivery
  - Common audio file formats include:
    - .wav
    - .mp3
    - .aiff
    - .acc
    - .midi
    - .m4a

Multimedia Elements

- Video: Begins as a continuous stream of visual information, which is then broken into separate images (frames) when the video is recorded
  - Can require a substantial amount of storage space
  - Video data, like audio data, is usually compressed
  - Streaming video is used for large files
  - Common video file formats include:
    - .avi
    - .mp2
    - .mov
    - .wmv
    - .mp4

Quick Quiz

1. The most common file format for Web page photographs is ________________________.
   a. GIF
   b. JPEG
   c. PNG
2. True or False: Delivery speed is one potential disadvantage of using Web-based multimedia.
3. A small image that is linked to a larger version of the same image is called a(n) ________________________.

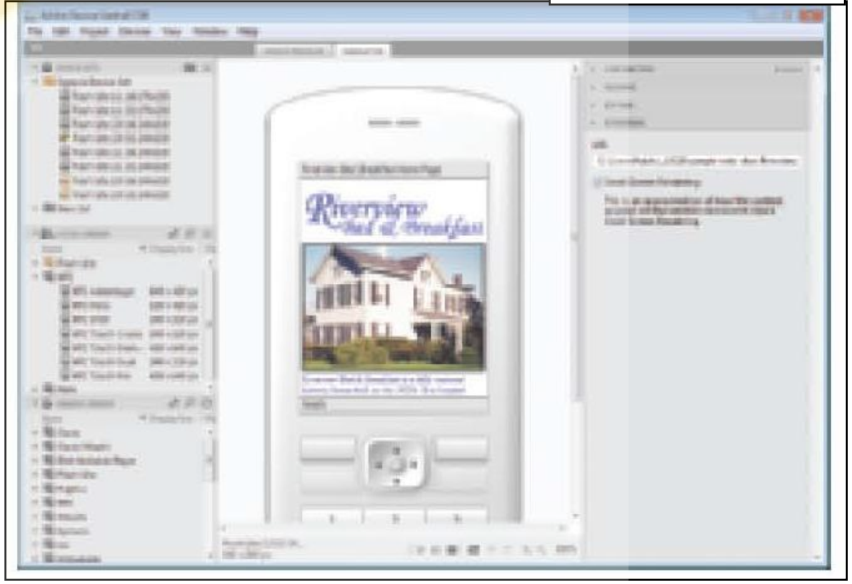
Answers:

1) b; 2) True; 3) thumbnail image

Multimedia Web Site Design

- Web site design: The process of planning what a Web site will look like and how it will function
  - Good planning pays off in the long run
- Basic design principles:
  - Users like interesting and exciting applications
  - Users have little patience with slow-to-load or hard-to-use applications
  - Plan for all needed delivery methods and devices

FIGURE 10-10
Testing on multiple platforms. Emulators (stand-alone or built into Web development programs) allow you to test your multimedia content on a variety of devices.



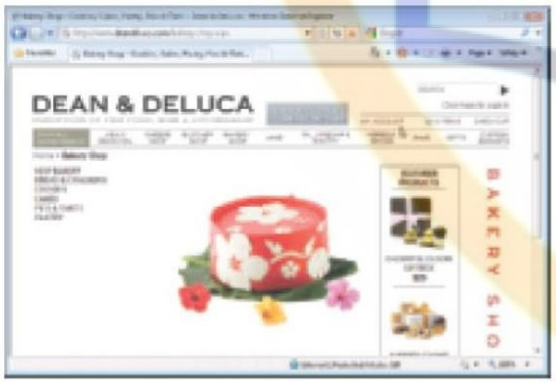
Multimedia Web Site Design

- Careful consideration should be given to:
  - Features that require a specific browser
  - Features that require little used plug-ins
  - The size of the page content
    - Different devices, browsers, and screen resolutions affect how Web pages display
  - High-bandwidth items
    - Watch image file size (use thumbnails)
    - Use links to audio, video, and other high-bandwidth items
    - Use streaming audio and video

Multimedia Web Site Design

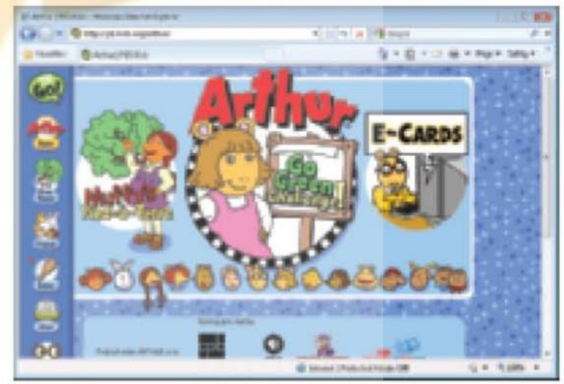
- Determining the intended audience and objectives
  - One of the first steps in designing a multimedia application or Web site
  - Objectives of the site affect its content
  - Intended audience affects the appearance (such as the style, graphics, fonts, and colors) of the site
  - Once the objectives and audience have been identified, you should have a good idea of the main topics to be included in the site
  - If the needed content is still unclear, rethink your audience and objectives and don’t go further in the process until it becomes clear

Multimedia Web Site Design



BOLD AND CLEAN
Shopping sites often use bold colors and crisp typefaces to give the site a contemporary, but rich and uncluttered, feel.

CONSERVATIVE
Many business sites use a conservative appearance to match th conservative business image.



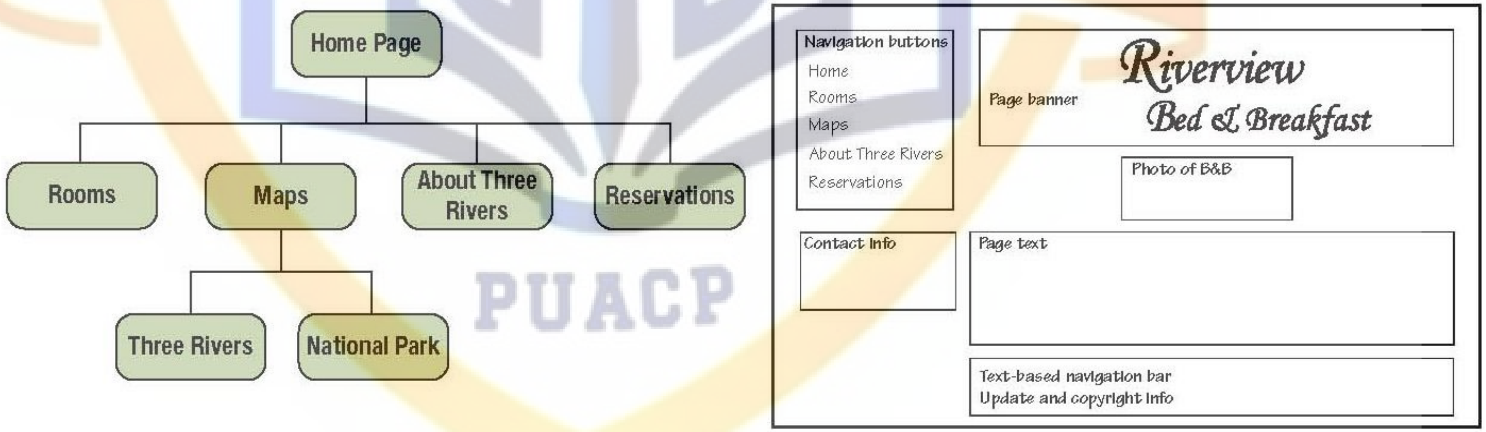
WHIMSICAL AND FUN
Sites catering to young people often have an especially friendly look, sporting bright graphics and large fanciful typefaces.

FIGURE 10-11
The intended audience affects Web site design.

Multimedia Web Site Design

- Flowcharts: Used during the Web design process to illustrate how the pages in a Web site relate to one another
- Page layouts: A sketch of a Web page often developed during the Web design process to illustrate the basic layouts of the home page and the rest of the pages on a Web site
  - Typically one for the home page and one for the rest of the pages on the site
- Storyboards: An ordered series of sketches that can be developed during the design process of an animated sequence or other multimedia component that illustrates what each page or screen will look like

Multimedia Web Site Design



FLOWCHARTS
Describe the logical organization of the site. Each box represents a separate Web page.

PAGE LAYOUTS
Illustrate the basic design and navigational structure of a Web site. There are typically two layouts—one for the home page (shown here) and one for all other pages on the site.

FIGURE 10-12
Web site flowcharts and page layouts.
A sample flowchart and page layout for a bed and breakfast Web site are shown here.

Multimedia Web Site Design

- Navigational design considerations
  - Users should be able to get to most pages on the site within three mouse clicks
  - Navigational items should be placed in the same location on every page
  - Each page should have a link to the home page of the site
  - Long Web pages:
    - Consider breaking into several pages
    - Include link to view or download entire document
    - Use table of contents and links to top of page

Multimedia Web Site Design

- Navigational tools include:
  - Drop-down menus
  - Site maps
  - Search boxes
  - Text-based hyperlinks and navigation bars
  - Menu tabs
  - Image-based navigation bars
  - Image maps
  - Hyperlinks that show more options when pointed to

Multimedia Web Site Design

FIGURE 10-13
Navigational tools.
A wide variety of navigational tools exists to help make Web sites easy to use.

Multimedia Web Site Design

- Access considerations:
  - Device compatibility: Web pages display differently on different devices
  - Compatibility with assistive technology
    - Alternate text
    - Meaningful hyperlinks
    - ADA compliance
  - Low readers or non-English speakers

FIGURE 10-14 Web pages display differently on different devices.

Multimedia Web Site Design

FIGURE 10-15
Some Web page characteristics that are compatible with assistive technology.

Quick Quiz

1. Which of the following is most often used to illustrate what a Web page will look like?
   a. Flowchart
   b. Storyboard
   c. Page layout
2. True or False: In order for a screen reading program to identify an image-based hyperlink, alternative text must be assigned to that image.
3. A Web page that contains links to all the main pages on a Web site is called a(n) ___________________________.

Answers:

1) c; 2) True; 3) site map

Multimedia Web Site Development

- Web site development: The process of creating, testing, publishing, and maintaining a Web site
  - Occurs after the site is designed
  - Can be performed in-house or outsourced
  - Three basic steps
    - Creating the multimedia elements
    - Creating the Web site
    - Testing, publishing, and maintaining the site

Multimedia Web Site Development

- Creating the multimedia elements
  - Usually several different software programs are used, such as:
    - Image editing software
    - Animation software
    - Audio editing software
    - Video editing software
- Each element should be saved in the appropriate size, resolution, and file format

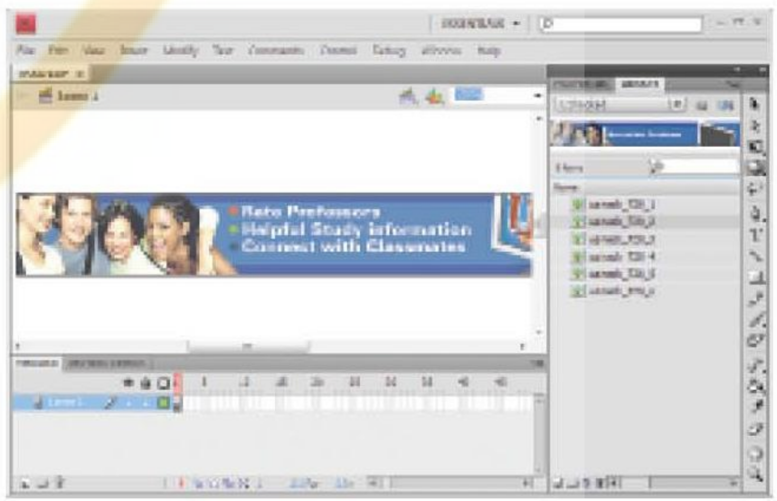


FIGURE 10-16
Web page banner being created in Adobe Flash.

Multimedia Web Site Development

- Creating the Web site
  - Often a markup language is used
  - Markup language: A language that uses symbols or tags to describe what a document should look like when it is displayed in a Web browser
  - Hypertext Markup Language (HTML): The original markup language
    - Uses HTML tags to indicate where effects and elements belong in the Web page
    - Some tags are paired
    - The computer and browser being used still determines exactly how the Web page will display

HTML

- HTML tags are used to:
  - Identify where elements (graphics, animations, video clips, etc.) should appear on the page
  - Assign a title to the page
  - Mark end of paragraphs
  - Specify the layout of tables and frames
  - Identify keywords and other meta tags associated with the page

HTML

TAG	PURPOSE
<html></html>	Marks the beginning and end of an HTML document.
<head></head>	Marks the head section which contains the page title and meta tags.
<title></title>	Marks the title of the Web page.
<body></body>	Defines attributes of an HTML Web document, such as background color, background image, text color, margins, etc.
to	Formats headings larger or smaller than the regular text in the document; H1 is the largest text.
<img>	Indicates an image file to be inserted; attributes included within this tag specify the image filename, display size, alternative text, border, etc.
<a>	Creates a hyperlink.
	Bolds text.
<i></i>	Italicizes text.
<center></center>	Centers text.
<hr>	Inserts a horizontal rule.
 	Inserts a line break (new line within the same paragraph).
<p>	Inserts a paragraph break (starts a new paragraph).

FIGURE 10-17
Sample HTML tags.

HTML

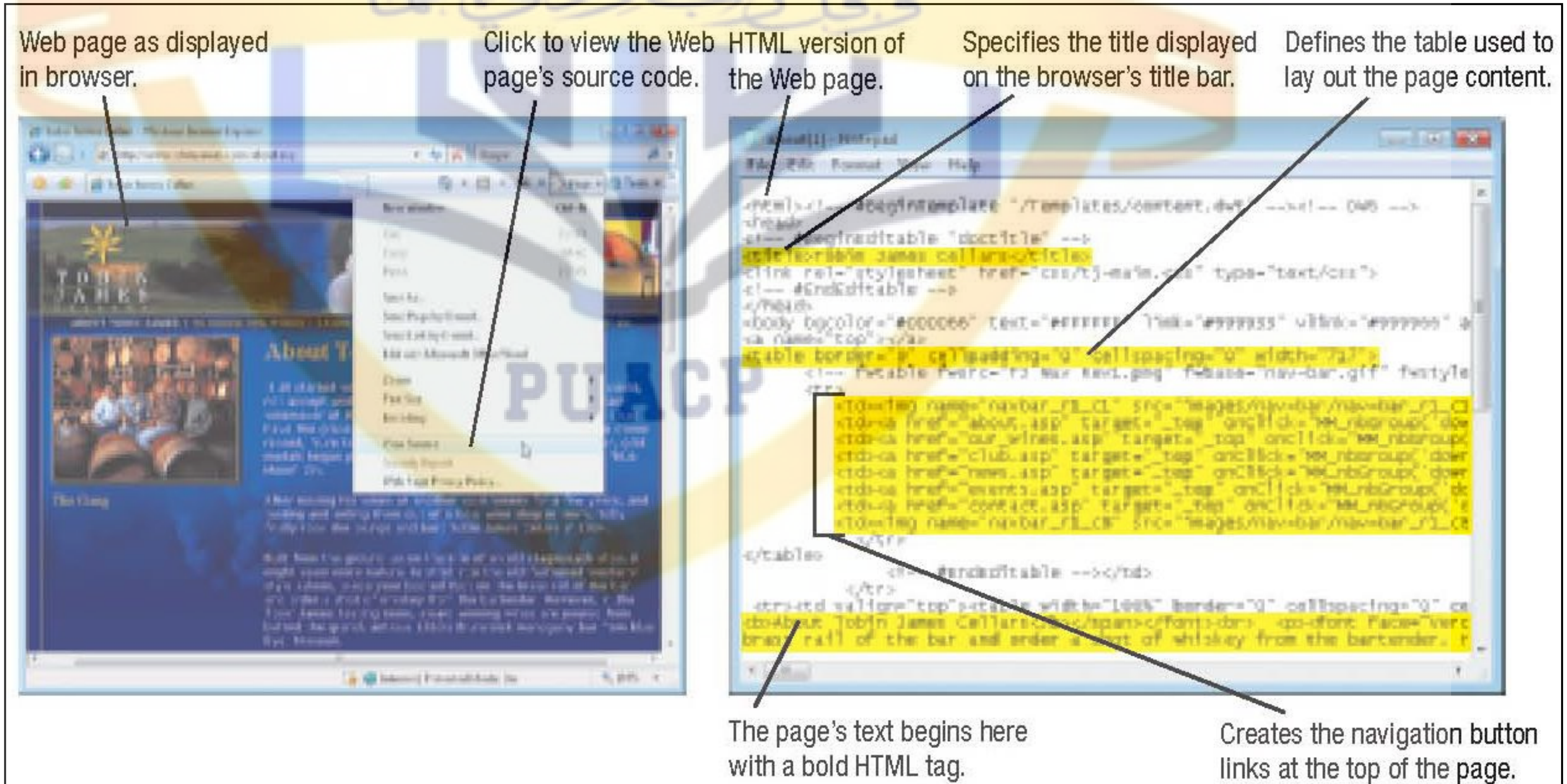


FIGURE 10-18
An HTML Web page and its corresponding source code.

Multimedia Web Site Development

- Extensible Markup Language (XML): A set of rules for exchanging data over the Web
  - Addresses the content but not the formatting
  - Uses XML tags to identify data
  - Allows data to be extracted and reused as needed
- Extensible Hypertext Markup Language (XHTML): A newer version of HTML based on XML
  - Controls the appearance and format of a Web page like HTML
  - Stricter rules than HTML
  - Beginning to replace HTML

XHTML

- XHTML tags: Used for the same purposes as HTML tags, but stricter:
  - All attribute values must be in quotation marks
  - Tags are lowercase (case-sensitive)
  - Tags must be closed
    - <p> and </p> or <p />
  - Tags must be in proper order
- Main sections of XHTML Web page
  - Declaration statement with XHTML standard used
  - Head statement with title and meta tags
  - Body of the Web page

XHTML

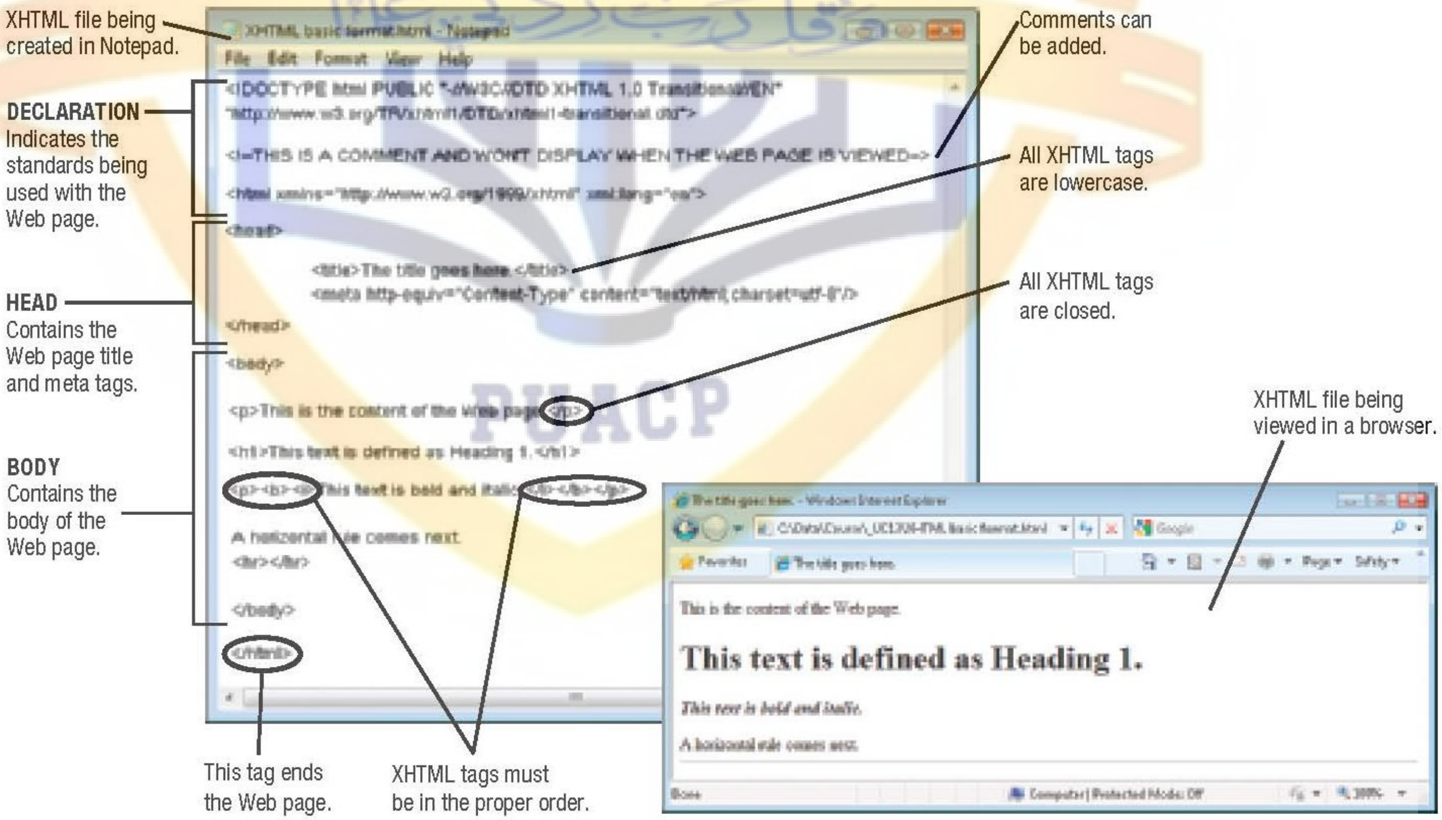


FIGURE 10-19
A sample XHTML Web page source code and corresponding Web page.

Multimedia Web Site Development

- HTML5: Newest version under development
  - Designed to replace current versions of HTML and XHTML
- Cascading Style Sheets (CSS): Used to specify the styles used with a Web page or an entire Web site
  - Internal or external style sheets
  - Pages link to style sheet, more efficient
- Wireless Markup Language (WML): Used to create Web pages to be displayed on WAP-enabled devices, such as smart phones

Multimedia Web Site Development

- Scripting language: Often used for dynamic content
  - Allows the inclusion of scripts (instructions) in the Web page code
  - JavaScript (resembles the Java programming language)
  - VBScript (based on Microsoft's Visual Basic programming language)
  - Perl (used to write CGI scripts to process data input via a Web page)
- AJAX: Creates faster, more efficient interactive Web applications
  - Only requests new data from the server, not the entire Web page, when the page is updated

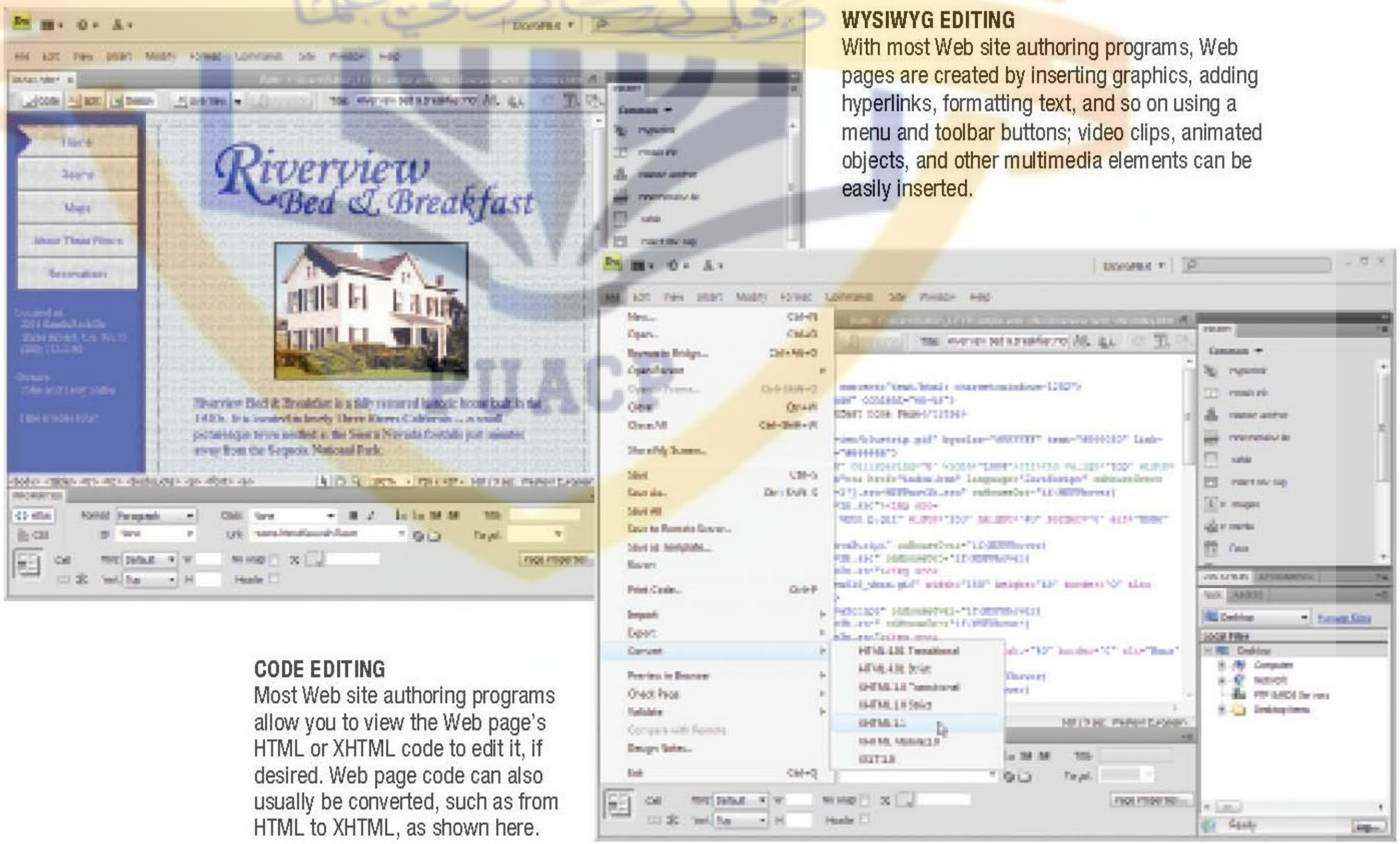
Multimedia Web Site Development

- Other content development tools
  - ActiveX: A set of controls that can be used to create interactive Web pages
    - Extends OLE to integrate content from two or more programs
    - Allows a variety of types of Windows files to be viewed via Web pages
  - Virtual Reality Modeling Language (VRML): A language used to create 3D Web pages
    - Successor is X3D

Multimedia Web Site Development

- Web site authoring software: Used to create Web pages and entire Web sites (Dreamweaver, Expression Web)
  - Toolbar buttons, menus, etc. are to create and format the page
  - The appropriate HTML statements are automatically generated
  - Allows you to create an entire cohesive Web site, not just individual pages
  - Allows you to easily include:
    - Forms and database connectivity your visitors
  - Often include tests for broken links & accessibility tests

Multimedia Web Site Development



WYSIWYG EDITING
With most Web site authoring programs, Web pages are created by inserting graphics, adding hyperlinks, formatting text, and so on using a menu and toolbar buttons; video clips, animated objects, and other multimedia elements can be easily inserted.

CODE EDITING
Most Web site authoring programs allow you to view the Web page's HTML or XHTML code to edit it, if desired. Web page code can also usually be converted, such as from HTML to XHTML, as shown here.

FIGURE 10-20
The Dreamweaver Web site authoring program.

Multimedia Web Site Development

- Testing, publishing, and maintaining the site
  - All hyperlinks should be clicked to ensure they take the user to the proper location
  - Every possible action that could take place with an animated element should be tested
  - Proofread each page or screen carefully
  - Consider a “stress test”
  - Update content and check links on a regular basis
  - XHTML Web pages can be checked with an XHTML validator

Multimedia Web Site Development

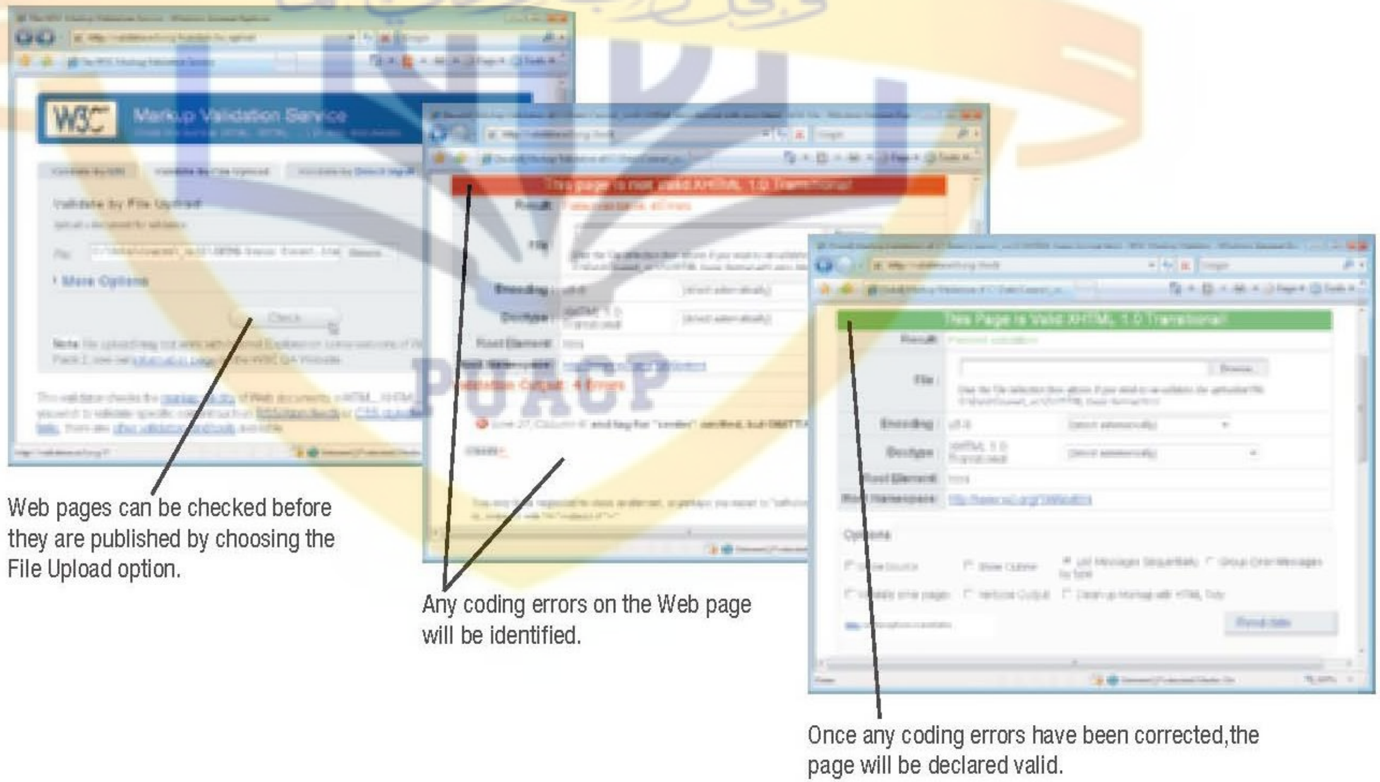


FIGURE 10-21
Validating an XHMTL Web page. This Web site (validator.w3.org) can be used to validate XHTML Web pages.

The Future of Web-Based Media

- Web-based multimedia will be more exciting and more embedded into everyday events
- Web-based multimedia and home entertainment devices will continue to converge
  - Allow seamless access to desired content on the user’s device
- Technology will evolve to support mobile multimedia
- Usage of multimedia applications that involve user generated content will continue to grow

Quick Quiz

1. Which of the following markup languages is most often used to create Web pages?
   a. HTML
   b. JavaScript
   c. WML
2. True or False: Web site authoring software can typically be used to create all of the Web pages on a site, including adding animated elements, video clips, etc.
3. The HTML code ___________________________ would begin to bold Web page text.

Answers:

1) a; 2) True; 3) <b>

Summary

- What Is Web-Based Multimedia?
- Multimedia Elements
- Multimedia Web Site Design
- Multimedia Web Site Development
- The Future of Web-Based Multimedia